GENERAL ELECTRIC

EV-1 SCR CONTROL

OPERATING INSTRUCTIONS
SYSTEM ANALYZER

(For 12-to 84-volt Systems)

TEST	PROCEDURE	RESULT OR ACTION
ANALYZER Shipped less batteries. (Quantity 4, AA Penlite, alkalyne type suggested.) After use return all switches to OFF. Do not store analyzer for long period of time with batteries installed.	1. SW1-TEST, SW2-OFF, SW3-OFF. Push LAMP TEST button.	1. All four lamps must light. If green lamp remains off and red lamps are on, replace green lamp. If red lamps are off, replace batteries. If all red lamps are not off or on as required, replace analyzer.
PREPARATION OF TRUCK FOR TESTING	**Disconnect truck battery. Raise and block drive wheels off floor.**	
TRUCK BATTERY	1. Check truck battery specific gravity.	1. Recharge battery if specific gravity is low.
PREPARATION FOR ELECTRICAL SYSTEM TESTS **All tests must be done sequentially.** **(Be sure truck battery is disconnected.)**	**Discharge commutating capacitor. Remove control card and install analyzer. See GEK-40724 Table 4A for card removal instructions. Check all fuses. Check all contactors visually for welded tips.**	
In all following tests, if component shows defective, first check associated wiring continuity before replacing component. Refer to GEK-40724 Table 4 for additional component checks where required, or Table 5 for component replacement instructions. After replacing components found defective by EV-1 Analyzer, re-run all electrical system tests with the Analyzer to insure the control is operational before driving truck. **If all electrical system tests are good, replace card. If control still misoperates, trouble shoot per instructions in GEK-40724. If a faulty component is then detected, recheck analyzer and replace analyzer if it does not detect faulty component.**		
1 REC	1. SW1-1 REC, SW2-OFF, SW3-OFF. 2. Depress and release 1 REC button.	1. Green lamp should be off. 2. Green lamp should light and remain on. If either condition is not met, replace 1 REC.
2 REC	1. SW1-2 REC, SW2-OFF, SW3-OFF. 2. Depress and release 2 REC button.	1. Green lamp should be off. 2. Green lamp should light and remain on. If either condition is not met, check 22 REC. If 22 REC is good, remove 2 REC power cathode wire and repeat test. If defective, replace 2 REC.
5 REC	1. SW1-5 REC, SW2-OFF, SW3-OFF. 2. Depress and release 5 REC button.	1. Green lamp should be off. 2. Green lamp should light and remain on. If either condition is not met, check 25 REC. If 25 REC is good, remove 5 REC power cathode wire and repeat test. If defective, replace 5 REC.
ELECTRICAL TESTS REQUIRING BATTERY POWER.	**Return all switches to OFF. Connect truck battery. Drive wheels must be off the floor.**	
BATTERY POLARITY	1. SW1-TEST, SW2-OFF, SW3-OFF.	1. All four lamps, should stabilize to the off condition within 5 seconds. If all red lamps are on, polarity is reversed. If capacitor lamp remains on, replace capacitor(s).
3 REC	1. SW1-TEST, SW2-OFF, SW3-OFF. Push 3 REC button and then release.	1. 3 REC lamp should light only when the button is depressed. If lamp does not light, check main power fuse and 23 FIL. If good, replace 3 REC.
CAPACITOR(S)	1. SW1-TEST, SW2-OFF, SW3-OFF. 2. Push CAP button and release.	1. If lamp lights, replace capacitor(s). 2. CAP lamp should come on and remain on approximately 1 second per capacitor. If not, replace capacitor(s).
PREPARATION FOR TESTING OF OPERATOR CONTROL SWITCHES, SYSTEM SWITCHES, AND CARD WIRING	**Person performing test should be on vehicle. Any switch is considered to be shorted, if it reads 20K ohms or less on R x 10,000 scale with battery disconnected and switch mechanically open.**	
KEY SWITCH AND CONTROL FUSE	1. SW1-TEST, SW2-KEY, SW3-OFF. 2. Turn on key.	1. Switch lamp should be off. If on, check for short. 2. Switch lamp should light. If not, check for open.
SEAT SWITCH	1. SW1-TEST, SW2-SEAT, SW3-OFF, key on. 2. Close seat switch.	1. Switch lamp should be off. If on, check for short. 2. Switch lamp should light. If not, check for open.
START SWITCH	1. SW1-TEST, SW3-BR/ST, SW3-OFF, key on, seat switch closed. 2. Depress accelerator.	1. Switch lamp should be off. If on, check for short. 2. Switch lamp should light. If not, check for open or open brake switch.
BRAKE SWITCH	1. Maintain above conditions and step on brake.	1. Switch lamp should go off. If not, check for short.

* Trademark of General Electric Company.

TEST	PROCEDURE	RESULT OR ACTION
DIRECTIONAL SWITCH (FORWARD)	1 SW1-TEST, SW2-FOR , SW3-OFF, key on, seat switch closed, brake switch closed, depress accelerator, and move directional switch to NEUTRAL 2 Move directional switch to FORWARD (See Note below) 3 Move directional switch to REVERSE (See Note below)	1 Switch lamp should be off. If on check for short 2. Switch lamp should light If not, check for open 3 Switch lamp should not light. If it does, check PMT driver terminal 3 to 4 for short If OK, replace hour meter diode (26 REC).
DIRECTIONAL SWITCH (REVERSE)	1. Interchange Steps 2 and 3 above and repeat	1 Same as above.

Note: If selection of directional switch to either F or R activates pump contactor, replace hour meter diode (26 REC).

CONTACTOR AND MOTOR TESTS	Truck drive wheels must be off the floor.	
FORWARD CONTACTOR, MOTOR CIRCUIT CONTINUITY, AND 3 REC OPEN	1. SW1-TEST, SW2-OFF, SW3-F/R, key on, seat switch closed, brake switch closed, move directional switch to NEUTRAL Push 3 REC button. 2 SW3-OFF Depress accelerator. Move directional switch to FORWARD. 3 SW3-F/R 4. Push 3 REC button.	1 3 REC lamp lights If not, replace 3 REC. 2 If forward contactor picks up, replace PMT driver 3. If contactor does not pick up, check coil and continuity of wiring If good, replace PMT driver. 4 3 REC lamp should remain off. If not check contactor for open tips Also check motor field and armature (should read less than 5 ohms with VOM on R x 1 scale)
REVERSE CONTACTOR AND MOTOR CIRCUIT CONTINUITY	1 Same as above except directional switch in REVERSE.	1 Same as above except for reverse contactor
FW CONTACTOR AND DRIVER	1 SW1-TEST, SW2-OFF, SW3-OFF 2 SW3-FW.	1. If FW picks up, replace FW driver 2. FW contactor should pick up. If not, check coil. If good, replace FW driver
1A CONTACTOR AND DRIVER	1 SW1-TEST, SW2-OFF, SW3-OFF 2 SW3-1A.	1. If 1A picks up, replace 1A driver. 2. 1A contactor should pick up. If not, check coil. If good, replace 1A driver. If 1A will not pick up under actual vehicle operation, check continuity of (VIO)wire from T2 to control plug.
B CONTACTOR AND DRIVER	1. SW1-TEST, SW2-OFF, SW3-OFF Close key and seat switch (plus other switches required to put battery positive to B card and coil) 2. SW3-D/B	1 B contactor should pick up If not, check coil. If good replace B card. 2 B contactor should drop out when SW3 is moved to D/B. If not, replace B card.
D (DUAL MOTOR) CONTACTOR AND DRIVER	1 SW1-TEST, SW2-OFF, SW3-OFF. (Jumper PMT driver terminals 2 to 4 and 3 to 4) 2 SW3-D/B 3 Key on, seat switch closed, brake switch closed, depress accelerator and move direction switch to FORWARD. Open each turn switch 4 Repeat Steps 2 and 3 for directional switch REVERSE.	1. If D contactor picks up, replace D card. 2 D contactor should pick up. If not, check D coil, 33 REC (if used), diodes, turn switches. If OK, replace D card. 3 D contactor and associated F contactors should de-energize If not, check for shorted turn switch If OK, replace D card 4. Same results as Steps 2 and 3.

DISCONNECT BATTERY BEFORE PERFORMING THE FOLLOWING THE FOLLOWING TESTS.

ACCELERATOR POT	SW1-OFF, SW2-OFF, SW3-OFF Connect VOM (R x 100 scale) into POT test jacks.	Resistance should be 1000 to 1200 ohms with accelerator fully depressed and 6000 to 7000 ohms released. If not, adjust accelerator or replace pot For reading directly across pot without analyzer, range is 0-200 ohms min and 5000-6000 ohms max
ACCELERATOR 1A SWITCH	SW1-OFF, SW2-OFF, SW3-OFF Connect VOM (R x 1 scale) into 1A SW test jacks.	Resistance should be less than 10 ohms with accelerator fully depressed. If not, check wiring and switch adjustment or replace 1A switch
THERMAL PROTECTOR	SW1-OFF, SW2-OFF, SW3-OFF. Connect VOM (R x 1 scale) into T. PROT test jacks.	Resistance should be less than 200 ohms at room temperature If not, check for loose connections or replace.
CURRENT SENSOR	SW1-OFF, SW2-OFF, SW3-OFF, Connect VOM (R x 1 scale) into SENSOR test jacks.	Resistance should be less than 1 ohm. If not, check for loose connections or replace.
POSITIVE OR NEGATIVE GROUND LEAKAGE	SW1-OFF, SW2-OFF, SW3-OFF Connect VOM (R x 10,000 scale) between POS GND test jack and truck frame or NEG GND test jack and truck frame.	Resistance should be greater than 20K ohms. If not, trouble shoot for ground fault.
4 REC	Cannot check with analyzer. Disconnect 4 REC pigtail. With VOM on R x 1 scale, should read 7-12 ohms in the conducting direction (+ →|-) with POS on heat sink and NEG on pigtail. Reverse meter leads With VOM on R x 10,000 scale, should read 50,000 ohms or higher	